

1. Define a variable. Write the syntax to declare an integer variable `age` and initialize it to 20.
2. List the basic data types in C along with their format specifiers used in `printf/scanf`.
3. What is the difference between `int`, `float`, and `char` data types? Give one real-life example variable for each.
4. Write a C program to declare two integer variables `a` and `b`, take input from the user, and print their sum.
5. What is the output of the following code?

```
int x = 5;
printf("%d", x);
```

6. Write the correct format specifier for: (a) integer (b) float (c) character (d) double (e) string
7. Differentiate between `=` (assignment operator) and `==` (equality operator) with an example.
8. Write a program to check whether a number entered by the user is positive or negative using `if-else`.
9. What is the purpose of `scanf()`? Write its syntax to read an integer into variable `n`.
10. Evaluate the following expressions (assume `a = 10`, `b = 3`): (a) `a + b` (b) `a % b` (c) `a / b` (d) `a * b`
11. Write a program using `if` statement to check if a number is even or odd.

12. What will be the value of `c` after execution?

```
int a = 4, b = 2, c;
c = a > b;
```

13. Identify the type of operator used in each case: `+`, `&&`, `<=`, `++`, `=`.
14. Write a program to find the largest of two numbers using `if-else`.
15. Write a program to find the largest of three numbers using **only nested `if`** statements (an `if` inside another `if`, no `else if` ladder).
16. Write a program using `switch` statement that takes a number from 1–7 and prints the corresponding day name (Monday–Sunday).
17. What is the purpose of the `break` statement inside a `switch` block? Write a short example showing what happens if it is left out.

18. Write a menu-driven `switch` program that performs basic arithmetic (+, -, *, /) based on an operator character entered by the user.

19. Write a program using nested `if` to check whether a number is positive, and if positive, further check whether it is even or odd.